USDA FoodData Central

Food & Nutrient Analytics

MongoDB · Hadoop MapReduce · Pig · Hive

Flask dashboard

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50,000

foods tested

1,126,928

linked nutrient records

Turn food records into useful, checked answers

01 / DATA

What is in the data?

Food names, types and nutrient amounts from USDA's October 2024 CSV package.

02 / ANALYSIS

What do we calculate?

Food counts and nutrient totals, averages and observation counts, grouped by food type.

03 / COMPARISON

What do we compare?

Indexed vs unindexed searches; Hive Text vs RCFile query time and storage.

Full source: 2,063,876 foods and 26,848,323 food–nutrient records.

One dataset, two processing branches

USDA CSV ZIP

THEME 1

MongoDB

Ingest → index → aggregate → profile

THEMES 2 & 3

Prepared files in HDFS

MapReduce summaries

Pig cleaning → Hive Text + RCFile

Reconcile answers → save evidence → Flask dashboard

Store documents, summarize data, measure searches

INGESTION & ANALYTICS

From raw records to summaries

Raw collections preserve source records.

Linked nutrient data supports analytical pipelines.

QUERY PROFILING

Prove the effect of an index

Run the same lookup with a collection scan and an index; inspect executionStats.

$match → $group → $project · $unwind · $facet

Filter · summarize · choose fields · expand arrays · return multiple summaries

Distribute files and calculate nutrient summaries

HDFS FILE OPERATIONS

Shared distributed storage

mkdir: create folders

put: upload files

cat / get: view or retrieve

rm -r: delete the demo folder

PYTHON STREAMING ON YARN

Map → shuffle → reduce

Mapper emits nutrient groups and amounts.

Reducers calculate count, total and mean.

Results are checked against reference answers.

Development: master + two Docker workers. Lab architecture: three physical Ubuntu systems.

Clean once, compare two storage formats

01 / PIG

Clean the HDFS data

Load prepared records; keep usable, nonnegative nutrient amounts. Save cleaned records to HDFS.

02 / HIVE

Same data, two tables

Text: readable row data.

RCFile: binary column-oriented storage, using compression in this project.

03 / BENCHMARK

Same questions, checked answers

Run identical analytical queries. Verify matching answers; record query time and storage bytes.

Tables are reused across query repetitions; storage is measured separately from query time.

A dashboard that makes the work visible

RUN

Control the workflow

Select a development size or full dataset. Follow stage progress and technical logs.

VIEW

Explain each theme

See raw MongoDB samples, aggregation answers, HDFS outputs and Hive comparisons.

VERIFY

Show the actual stored data

Copy the command beside a result. Run it in a second terminal with that service running.

Saved results remain visible even when services are stopped. Live commands need the services running.

50,000-food development run passed

DATA PROCESSED

50,000 foods

1,126,928 linked nutrient records

477 nutrient definitions

INDEX PROFILE

50,000 → 1

Documents examined for the same lookup: collection scan vs index.

CORRECTNESS

Cross-system checks

Linked-subset answers passed reconciliation across the processing systems.

Outcome: a repeatable analytics workflow with visible results and measurable performance.

Scope: tested on one laptop with Docker workers; full-release analytics and the physical lab run are separate.

Source: fdc.nal.usda.gov/download-datasets/ · Code: github.com/Dhanush-singh-g/BDA-Assignment